

UST'-UMAL'TA, Russian Federation

WMO: 314740

Lat: 51.634N

Lon: 133.260E

Elev: 385

StdP: 96.79

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.1	-40.1	-45.1	0.0	-41.8	-43.3	0.1	-40.0	4.9	-10.1	3.1	-19.5	0.0	0	0.597

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	29.4	18.5	27.3	17.7	25.6	17.0	20.7	25.2	19.7	24.1	18.7	23.0	1.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	14.9	22.0	18.4	13.9	20.9	17.3	13.0	20.2	61.8	25.2	58.2	24.2	55.0	23.0	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.9	3.5	2.9	-44.1	32.9	2.5	1.8	-45.9	34.2	-47.3	35.2	-48.7	36.2	-50.5	37.5		
			-43.8	23.6	2.2	2.4	-45.4	25.3	-46.7	26.7	-47.9	28.0	-49.5	29.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.7	-28.1	-21.8	-11.0	0.6	9.5	15.9	19.2	16.7	9.6	-0.8	-16.3	-27.5
	DBStd	17.59	5.45	5.64	6.17	4.75	3.96	3.41	2.61	3.20	4.11	5.22	7.92	6.20
	HDD10.0	5410	1180	891	652	282	57	2	0	1	56	337	788	1163
	HDD18.3	7772	1438	1124	911	531	274	87	21	69	263	594	1038	1422
	CDD10.0	761	0	0	0	1	42	178	286	209	44	1	0	0
	CDD18.3	81	0	0	0	0	1	13	48	19	1	0	0	0
	CDH23.3	987	0	0	0	2	49	249	492	186	9	0	0	0
	CDH26.7	247	0	0	0	0	10	62	144	31	1	0	0	0
Wind	WSAvg	0.8	0.3	0.6	1.0	1.2	1.1	0.9	0.8	0.7	0.8	0.8	0.5	0.3
Precipitation	PrecAvg	753	7	8	16	36	87	107	151	145	98	59	30	17
	PrecMax	966	24	26	46	87	156	238	255	249	178	120	101	49
	PrecMin	590	2	0	1	8	34	28	9	65	36	22	7	5
	PrecStd	95	5	5	10	20	35	53	56	45	38	25	18	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.4	-3.3	6.9	20.6	27.9	31.2	32.8	29.4	24.6	16.9	2.2	-5.8
		MCWB	-8.6	-6.0	0.8	9.8	14.9	19.1	19.7	19.2	16.6	9.4	-1.2	-7.7
	2%	DB	-12.8	-6.6	3.8	15.6	24.3	28.2	30.2	27.2	21.5	11.8	0.1	-9.8
		MCWB	-13.8	-8.6	-1.1	6.9	12.5	17.0	19.3	18.3	14.3	5.9	-1.5	-10.9
	5%	DB	-16.5	-9.1	1.6	11.9	21.5	26.2	28.1	25.3	19.2	8.9	-1.9	-14.0
		MCWB	-17.3	-10.7	-2.3	4.4	10.9	16.1	18.5	17.7	13.0	4.5	-3.2	-14.6
	10%	DB	-19.6	-11.8	-0.6	8.9	18.7	24.0	26.1	23.6	16.9	6.5	-4.0	-18.0
		MCWB	-20.2	-13.2	-3.8	2.8	9.8	15.3	17.9	17.1	12.2	3.3	-5.3	-18.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.8	-5.7	1.3	10.0	15.4	20.6	23.1	21.5	18.0	10.0	0.1	-7.7
		MCDB	-6.4	-3.6	5.8	17.9	24.6	26.9	27.6	24.8	22.3	15.0	1.0	-5.9
	2%	WB	-13.7	-8.5	-0.5	7.3	13.6	18.7	21.2	20.1	15.7	7.0	-1.5	-11.0
		MCDB	-12.7	-6.7	3.2	15.2	20.8	25.1	25.9	23.9	19.2	9.9	0.0	-9.8
	5%	WB	-17.2	-10.7	-2.1	4.9	12.2	17.6	20.2	19.2	14.3	5.2	-3.3	-14.6
		MCDB	-16.5	-9.0	1.4	11.3	18.7	23.1	24.9	23.0	17.6	8.3	-1.9	-14.0
	10%	WB	-20.1	-13.1	-3.7	3.1	10.9	16.6	19.2	18.3	13.1	3.5	-5.3	-18.5
		MCDB	-19.6	-11.8	-0.6	8.1	17.1	21.5	23.6	21.8	16.0	6.1	-3.8	-18.0
Mean Daily Temperature Range	5% DB	MDBR	14.1	16.9	16.2	12.6	14.1	13.9	12.3	12.2	12.7	11.1	12.4	12.3
		MCDBR	14.2	16.5	15.3	17.7	20.3	18.0	16.8	15.1	16.3	14.8	10.5	11.2
		MCWBR	13.3	14.4	11.0	9.8	9.7	7.9	7.0	7.1	9.0	9.2	8.5	10.3
	5% WB	MCDBR	14.3	16.2	14.4	16.6	16.7	14.2	12.6	11.4	13.5	12.9	9.6	11.0
		MCWBR	13.3	14.1	10.3	9.4	8.7	7.1	6.7	6.4	8.4	8.6	8.0	10.2
Clear-Sky Solar Irradiance	taub	0.259	0.297	0.364	0.495	0.558	0.515	0.493	0.416	0.386	0.394	0.320	0.260	
	taud	2.124	2.039	1.892	1.727	1.759	1.958	2.074	2.304	2.298	2.059	2.032	2.081	
	Ebn at Noon	739	803	803	733	710	752	762	811	791	670	623	657	
	Edh at Noon	74	108	155	208	211	175	153	116	104	107	82	65	
All-Sky Solar Radiation	RadAvg	1.53	2.71	4.33	5.03	4.93	5.53	5.14	4.44	3.30	2.08	1.46	1.18	
	RadStd	0.04	0.07	0.18	0.49	0.47	0.58	0.36	0.38	0.34	0.16	0.12	0.04	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	+0.47	+0.61	N/A	N/A	N/A	N/A
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page